



# Shifts in the California Snow-to-Rain Ratio

Implications for Water Managers in a  
"Business as Usual" Future Emission Scenario

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## Why are the changes in the California Snow-to-Rain Ratio Important to Understand?

California is a state with frequent drought events and issues, making hydrological factors here much more pertinent than other US states. Climate change is expected to exacerbate these drought activities (Sokolo et. al., 2019), likely to negatively impact drinking water amounts, irrigation for agriculture, and other municipal uses. One way to understand how water supply may change over time is by looking at the snow-to-rain precipitation ratio, which describes the amount of snowfall precipitation compared to rainfall precipitation (also known as fluxes). Here I present my results, map products, and more, in order to understand how the snow-to-rain ratio may change with moderate climate change projections (SSP3-7.0). These results aim to further aid water

managers in making efficient and sustainable future hydrologic distribution strategies.

## Research Question

How will the snow-to-rain ratio change with SSP3-7.0 climate scenarios in California? And how will these findings aid in future water management decisions in California?

## Project Workflow

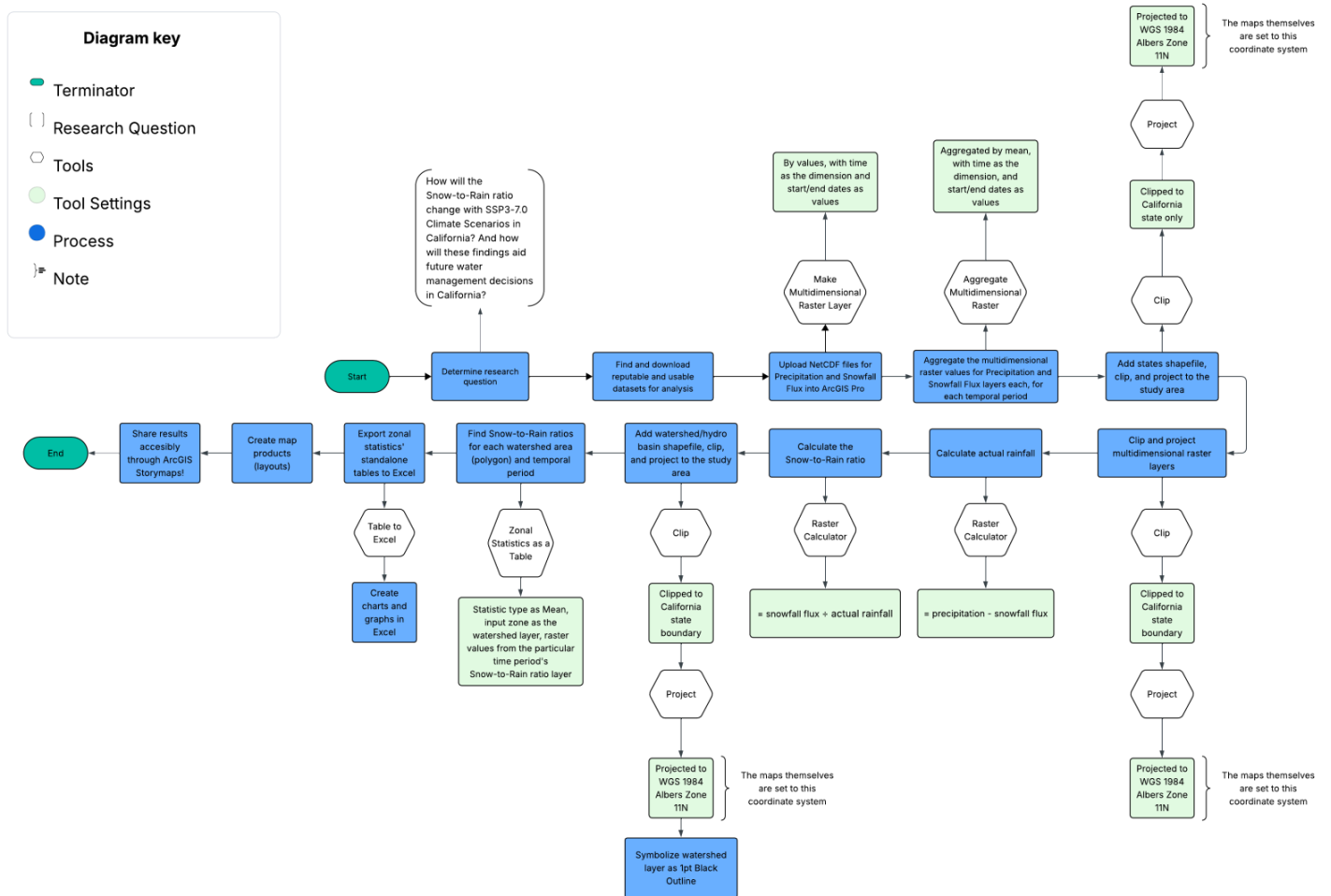


Figure 1. depicts the steps taken to complete my project and answer my research question. Tools, tool settings, processes, notes, my research question, and even the beginning/end of my process is symbolized with unique shapes (see Legend).

# Resulting Map Products

## Map Layouts

### Historical Data Results

#### Changes in the Snow-to-Rain Ratio Throughout Three Historical Time Periods in California: 1991-2000, 2001-2010, and 2011-2014

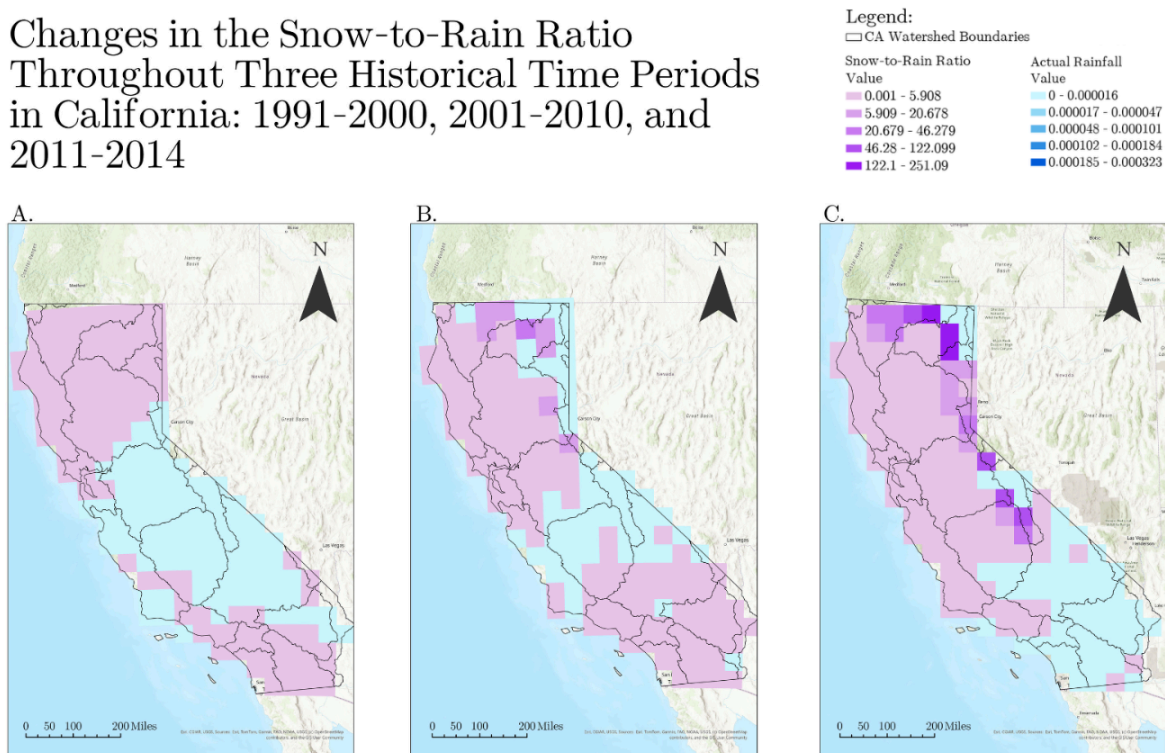


Figure 2. The three maps shown represent the changes in average snow-to-rain ratio for each historical time period in the study. Map A displays data results from 1991-2000, map B displays 2001-2010, and map C displays data results from 2011-2014. The Snow-to-Rain Ratio layer describes how much more snowfall flux there was compared to rainfall flux. Higher values and darker purple pixels show greater snowfall flux to rainfall flux. The Actual Rainfall layer was used to calculate the Snow-to-Rain Ratio layer, but is still displayed to accurately show precipitation activity. By looking at the three maps in sequence, it is clear that snowfall flux is increasing compared to rainfall flux. The two precipitation layers are overlaid by the state watersheds for further impact understanding.

### Future Data Results

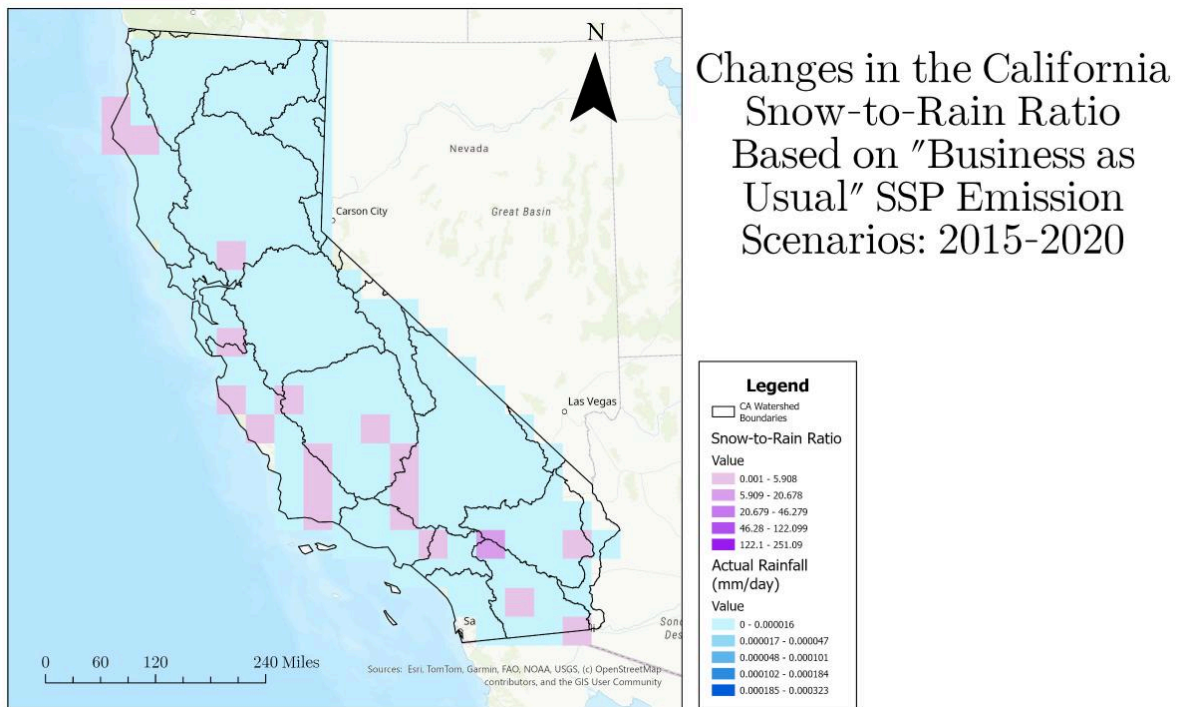


Figure 3. depicts the average snow-to-rain ratio by watershed in California during the 2015-2020 time period. Precipitation and snowfall flux data were taken from the SSP3-7.0 emission scenario, indicating steady levels of emissions without extremes. This map shows a decrease in snowfall flux compared to rainfall flux, indicating a time period with much less snowfall than in periods past.

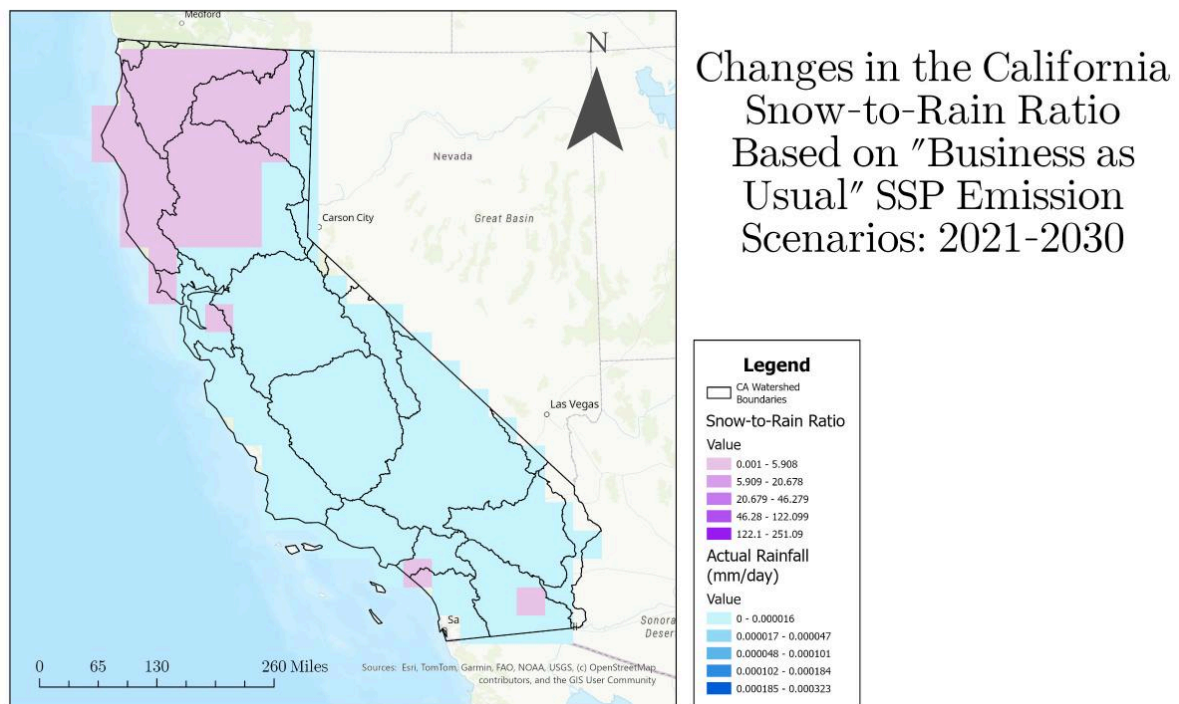


Figure 4. depicts the average snow-to-rain ratio by watershed in California during the 2021-2030 time period. Precipitation and snowfall flux data were taken from the SSP3-7.0 emission scenario, indicating steady levels of emissions without extremes. This map shows a change in distribution in the snow-to-rain ratio (purple pixels), displaying more snowfall activity in Northern California watersheds compared to southern or central California watersheds.

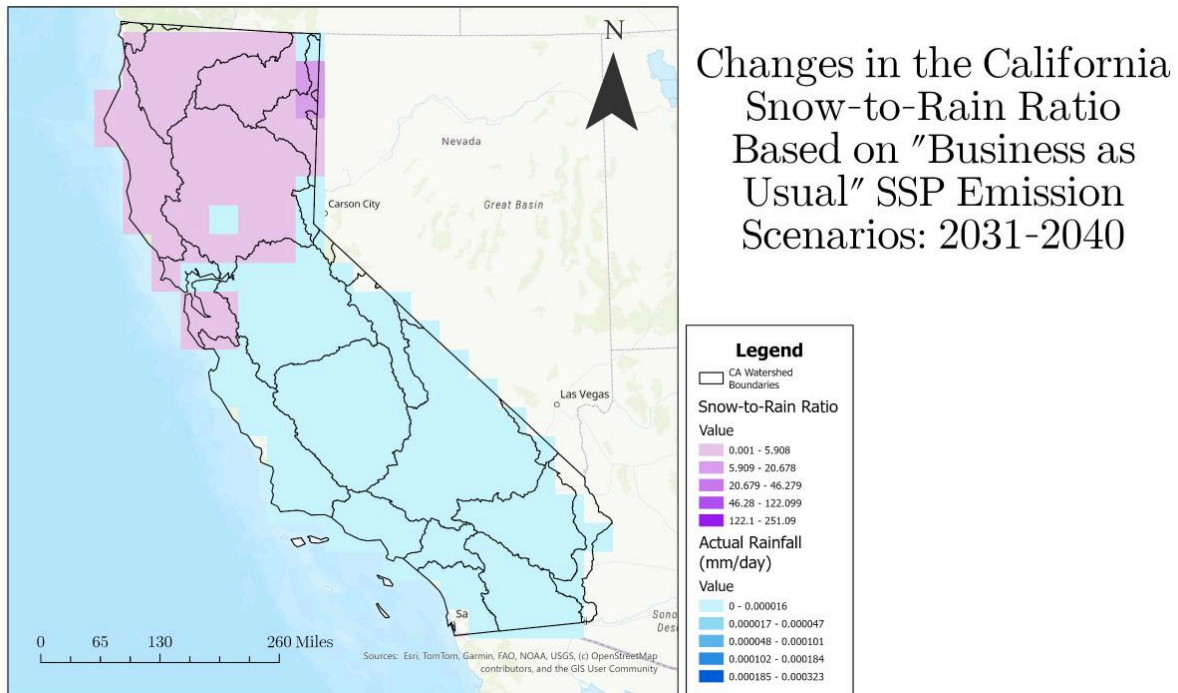


Figure 5. depicts the average snow-to-rain ratio by watershed in California during the 2031-2040 time period. Precipitation and snowfall flux data were taken from the SSP3-7.0 emission scenario, indicating steady levels of emissions without extremes. This map shows further distribution change in the snow-to-rain ratio (purple pixels). No snowfall flux was found in southern and central California watersheds, but greater snowfall fluxes than rainfall fluxes were found in northern California during this time period.

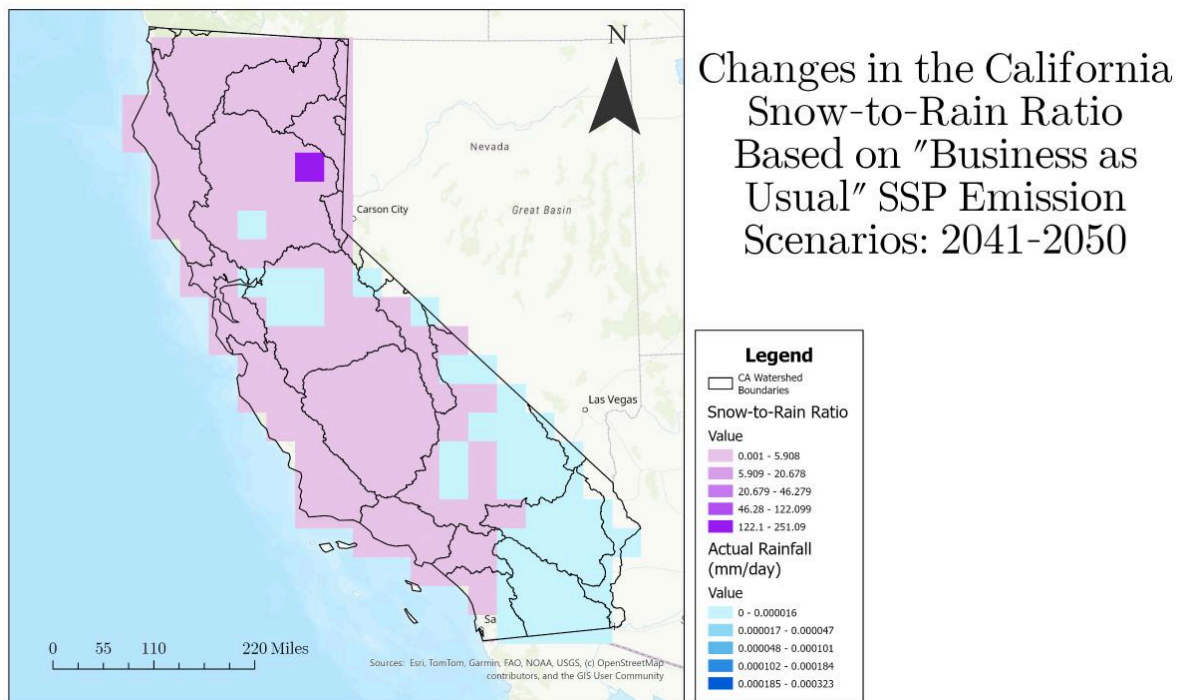


Figure 6. depicts the average snow-to-rain ratio by watershed in California during the 2041-2050 time period. Precipitation and snowfall flux data were taken from the SSP3-7.0 emission scenario, indicating steady levels of emissions without extremes. This map also shows more distribution change in the snow-to-rain ratio (purple pixels), indicating a more vast distribution across California. A greater snowfall to rainfall flux is indicated in the Lower Sacramento watershed. More snowfall to rainfall fluxes were also found in other central and southern California watersheds.



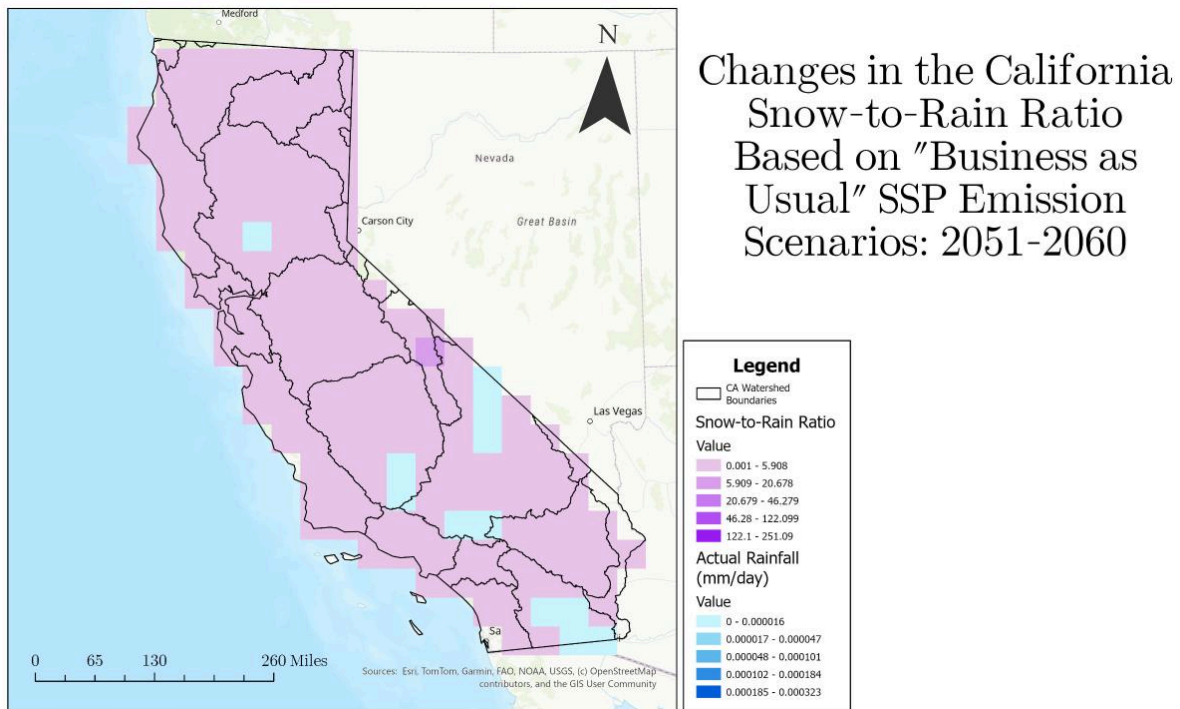


Figure 7. depicts the average snow-to-rain ratio by watershed in California during the 2051-2060 time period. Precipitation and snowfall flux data were taken from the SSP3-7.0 emission scenario, indicating steady levels of emissions without extremes. This map continues to show an increase in snowfall flux to rainfall flux from previous time periods across the state of California. The Snow-to-Rain Ratio layer here shows lower values indicating a much smaller difference in the ratio, but overall shows more snowfall than rainfall flux.

## Bar Charts

### Snow to Rain Flux Ratio by Watershed Basins in California

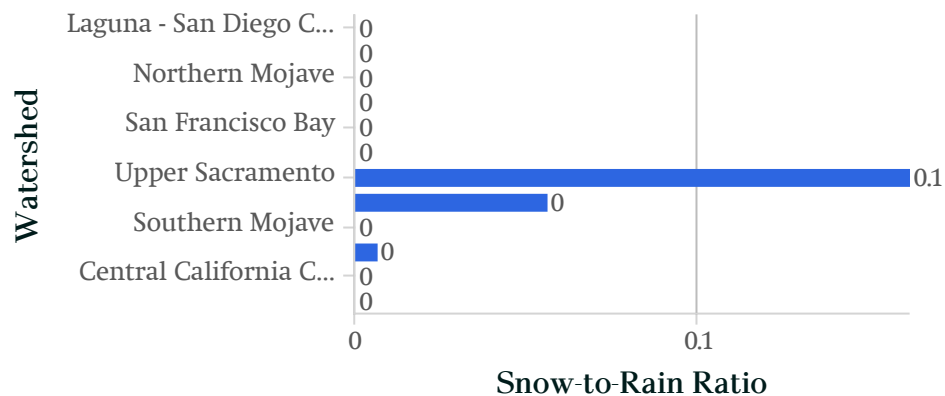


Figure 8. Shows historic 1991-2000 Snow-to-Rain ratios in California, averaged by each watershed in the state. Basically no watersheds had a great difference in the average fluxes of snow to rain precipitation during 1991-2000 the time period.

## Snow to Rain Flux Ratio by Watershed Basins in California 2001-2010

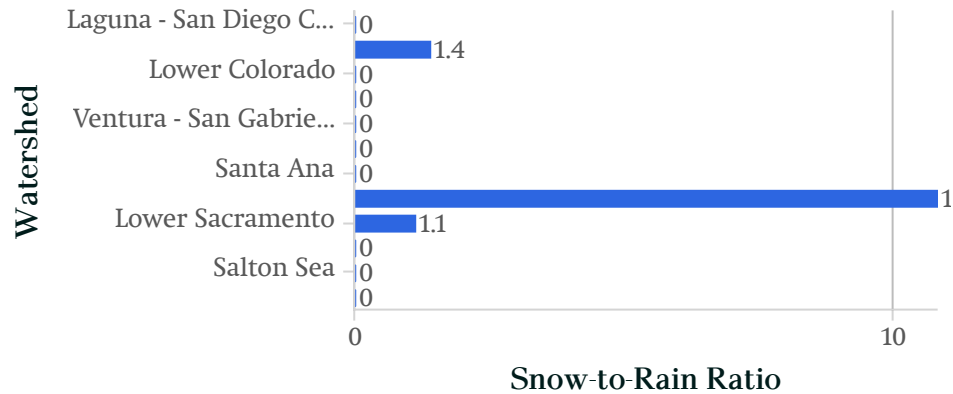


Figure 9. Shows historic 2001-2010 Snow-to-Rain ratios in California, averaged by each watershed in the state. Watersheds such as San Joaquin, Upper Sacramento, and Lower Sacramento showed greater amounts of snow to rain precipitation during 2001-2010 the time period.

For example, and for better understanding, the Upper Sacramento watershed snowfall flux was ~10.8 times greater than its rainfall flux.

## Snow to Rain Flux Ratio by Watershed Basins in California 2011-2014

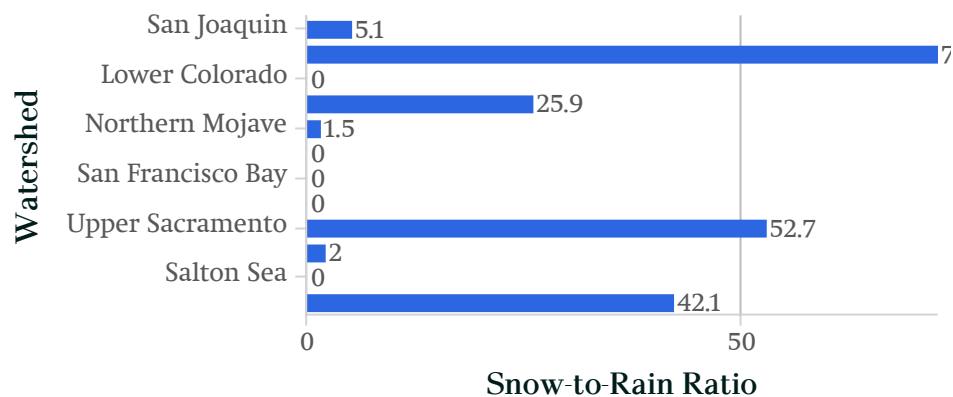


Figure 10. Shows historic 2011-2014 Snow-to-Rain ratios in California, averaged by each watershed in the state. Watersheds such as San Joaquin, Walker, Truckee, Upper Sacramento, and Mono - Owens Lake showed greater amounts of snow to rain precipitation during 2011-2014 the time period. The Walker watershed showed the largest difference in snow-to-rain ratio with a snowfall flux ~72 times greater than its rainfall flux.



## Snow to Rain Flux Ratio by Watershed Basins in California 2021-2030

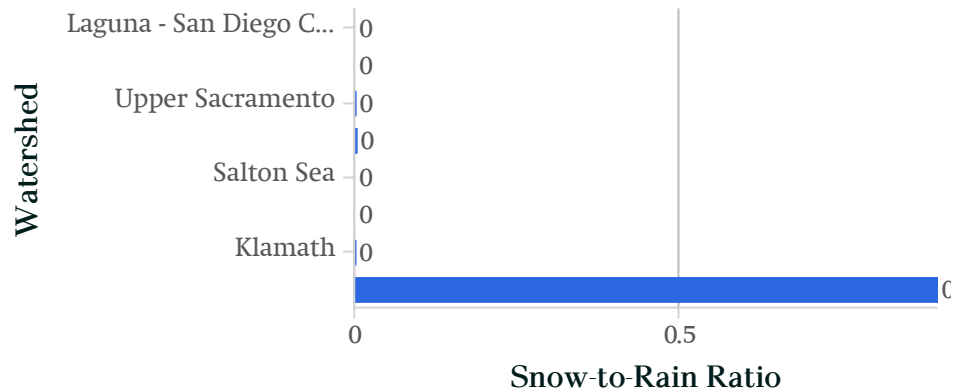


Figure 11. Shows Snow-to-Rain ratios in California from 2021-2030, averaged by each watershed in the state. Majority of watersheds during this period showed no difference in their average snow-to-rain ratio, except for Northern Lahontan which showed a resulting average snowfall flux 0.9 times greater than its rainfall flux. A few watersheds' (Lower Sacramento, Upper Sacramento, and Klamath) mean ratio values were rounded down to zero in the making of the chart (due to number of significant figures), but actually have very small mean values showing slightly great snowfall flux than rainfall flux. Lower Sacramento showed 0.0021, Upper Sacramento showed 0.0002, and Klamath showed 0.000021 times more snowfall flux than rainfall flux.

## Snow to Rain Flux Ratio by Watershed Basins in California 2031-2040

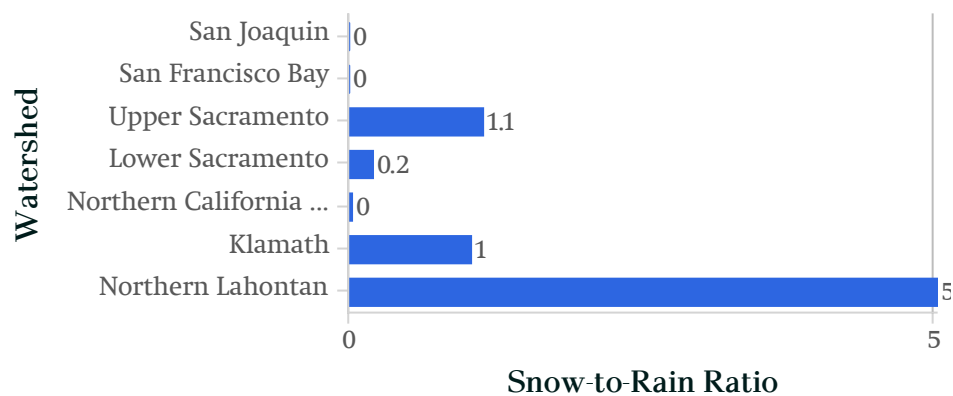


Figure 12. Shows Snow-to-Rain ratios in California from 2031-2040, averaged by each watershed in the state. The greatest difference in the snow-to-rain ratio was found in the Northern Lahontan watershed, with 5 times more snowfall than rainfall flux.

## Snow to Rain Flux Ratio by Watershed Basins in California 2041-2050

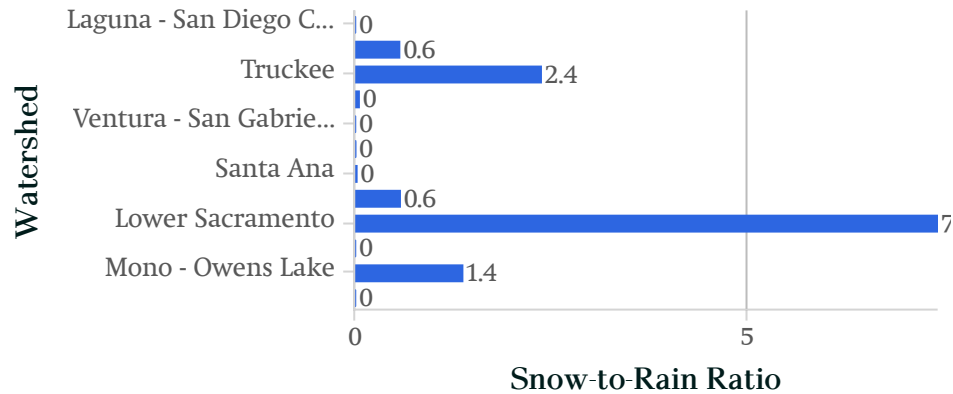


Figure 13. Shows Snow-to-Rain ratios in California from 2041-2050, averaged by each watershed in the state. The greatest difference in the snow-to-rain ratio was found in the Northern Lahontan watershed, with 5 times more snowfall than rainfall flux.

## Snow to Rain Flux Ratio by Watershed Basins in California 2051-2060

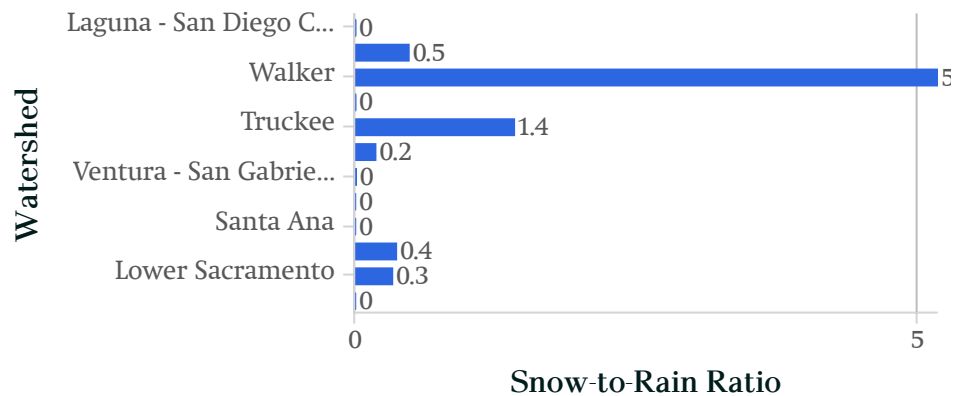


Figure 14. Shows Snow-to-Rain ratios in California from 2051-2060, averaged by each watershed in the state. The greatest difference in the snow-to-rain ratio was found in the Walker watershed, with a little over 5 times more snowfall than rainfall flux. The Truckee watershed also showed greater snowfall to rainfall flux on average during the 2051-2060 time period.

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## Discussion

### Real World Results

The historical map results show increasing amounts of snowfall flux compared to rainfall flux. The map products from the three historical periods represent times where California was more snow dominated, which was highest in the 2011-2014 period. However, from this point and through the future SSP3-7.0 scenario analyses, my study found a decrease in the snow-to-rain ratio. This is despite certain watersheds showing to have higher snowfall than rainfall flux, as certain mountainous air regions are more likely to produce more snow than rain compared to watersheds with different topography.

### Water Management Implications

According to the results, central and southern California watersheds showed the largest losses in their average snow-to-rain ratios. Watersheds such as Santa Ana, Ventura - San Gabriel Coastal, Laguna - San Diego Coastal, and others nearby should take greater attention from water managers and their water supply distribution strategies due to these losses. More adaptive and especially dynamic strategies will need to be implemented and likely altered between periods at these watersheds in order to keep a sustainable water supply for all of California's needs (environmental, agricultural, urban, etc.)

### Study Limitations

The pixel size for the precipitation and snowfall flux rasters were large, making it difficult to fully clip them to the study boundary. This also likely had an impact on the precision and spatial accuracy of the analysis, especially in the raster calculator. The time periods defined in this study are also very large, there could be finer details that are unseen due to the 5-10 year span of these periods.

### Future Work

1. Dynamic and adaptive water management strategies for central and southern California watersheds

2. Smaller scaled rasters for more precise analysis and clearer map products.
3. Smaller time scale for better detail and impact understanding.

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